CarbonPilot

An AI carbon decision engine. It turns fragmented operational data into prioritised, financially-aware decarbonisation actions — and every number it prints is traceable to a calculation engine, never to the language model.

Measure → Diagnose → Simulate → Act

The problem

- Companies can already measure their carbon. They still cannot decide what to do about it — the data exists, the decision does not.
- Reporting is a backward view; management asks what changes next, and what it will cost.
- Carbon and money are decided separately, and nobody can see the trade-off in one place.

The motivating finding

The same 4,304,333 kWh load costs **3,036 tCO_2e** on Indonesia's grid — rank 19 of 20 in JAPAC, coal 69% — and **399 tCO_2e** in New Zealand. Location is a decision variable, not a footnote.

The answer

DEMO FOOTPRINT	SCOPE 3
11,208 tCO_2e / yr	**61.1** %

- **Ingest.** CSV mapped to canonical activities; interpretation confirmed before anything is written.
- **Diagnose.** Robust median/MAD test finds the months that moved, then attributes drivers.
- **Simulate.** Interventions applied to the stored baseline; a constraint search explores combinations.
- **Act.** Ranked on carbon × cost × effort, weighted by factor confidence, tracked to an owner.

> One architectural rule: the language model never produces a carbon number.

Result, from the deployed instance

"Reduce our footprint 15% without increasing operating cost."

REDUCTION	CARBON SAVED	COST IMPACT	PASSES
33.8 %	**3,790** tCO_2e / yr	**−$775** K / yr	**342** of 2,098

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Because it is built to be trusted

EVIDENCE	VALUE
Behaviour-contract tests	`105`
Retrieval over a cited corpus	`BM25, stdlib`
Tool surface	`13 tools · forced first call`
Deploy shape	`1 process · static UI`
Languages	`EN / ID`
Smoke test	`fails on a dead key`

JAPAC by design

- Twenty regional grids benchmarked on one documented basis (Ember data, CC BY 4.0) — a public facility, not a slide.
- Bilingual EN / ID for the mid-market teams reporting platforms under-serve.

Honest boundaries

- No compliance reporting (CSRD / SB 253 / CDP) yet; curated point factors; linear scenario maths; no multi-tenancy — stated here rather than discovered by a judge.

> Carbon reporting tells you where you are. CarbonPilot tells you what to do next.

Proposal deck (PDF), demo video (3 min) and this summary accompany the working prototype. Repository public at submission. Google Cloud AI Builder Cup 2026 · submission deadline 18 October 2026.